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Continuous Behavioral Authentication for Mobile Devices Using an Adaptive LSTM-Based Intelligent Agent

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Abstract

Continuous authentication from smartphone inertial sensors must remain useful when a legitimate user's score distribution changes, yet an adaptive decision rule can also increase false acceptance or become vulnerable to profile poisoning. We evaluate a two-layer long short-term memory (LSTM) classifier and a lightweight agent that updates exponentially weighted first and second moments of accepted confidence scores. The study uses accelerometer and gyroscope data from 100 HMOG subjects. To prevent leakage, the protocol reads only the physical sensor columns, constructs non-overlapping sequences within individual sessions, keeps complete sessions disjoint between model fitting, validation, and test, and estimates normalization and initial operating points without test data. Linear SVM, random forest, and LSTM receive the same 10-second input and achieve undrifted-session macro equal error rates of 29.92% ± 0.80%, 24.05% ± 0.86%, and 21.87% ± 1.01%, respectively. At the validation-calibrated operating point, the frozen LSTM threshold and the EMA agent obtain HTERs of 23.19% ± 0.95% and 24.53% ± 1.28% in a balanced alternating zero-effort impostor stress test. At a final standardized-feature drift magnitude $\|\delta\|_2 = 2$, their HTERs are 24.89% ± 1.37% and 26.39% ± 2.12%. Thus the tested EMA rule does not improve the frozen threshold. These results provide a controlled ablation of the tested adaptive component. The contribution is a reproducible, session-disjoint evaluation of one prespecified EMA update rule, not a claim that adaptation is generally beneficial.

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Keywords: continuous authentication; behavioral biometrics; LSTM; adaptive threshold; inertial sensors; HMOG; concept drift

1. Introduction

One-time authentication cannot protect an unlocked smartphone after the device passes into another person's control. Continuous authentication (CA) instead evaluates behavioral signals throughout a session, often without explicit

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interaction [1, 2]. Accelerometers and gyroscopes are attractive for this purpose because they are widely available and capture user-specific handling and motion patterns [3, 4].

The difficulty is temporal change. A classifier trained during enrollment can become poorly calibrated as behavior and activity context change [5]. Retraining a deep model after every observation is impractical on a mobile device, whereas a score-level update is inexpensive. That convenience is not evidence of effectiveness: an update rule must reduce false rejection without exposing the operating point to impostors, and it must be evaluated on held-out data with all parameters fixed before test scoring.

We isolate the adaptation claim with a session-disjoint three-way protocol, an explicit EMA decision equation whose parameters are fixed before testing, and a paired frozen-LSTM versus LSTM+agent ablation at the same initial operating point. Genuine attempts are processed in session and within-session order in label-blind mixed streams. We report FAR, FRR, and HTER on undrifted sessions and under synthetic drift, while limiting conclusions to the tested protocol.

2. Related Work and Study Positioning

CA has a substantial literature. Broad surveys cover system components, modalities, metrics, privacy, and open deployment problems [2, 6, 7]. A contemporary smartphone-focused survey catalogs more than 140 sensor and behavioral-biometric approaches [4]. The present work therefore does not claim that continuous authentication, inertial sensing, LSTM classification, or EMA by itself is new.

HMOG introduced hand-movement, orientation, and grasp features for smartphone authentication [3]. Subsequent work applied augmentation, deep sensor models, and adaptive verification to mobile behavioral data [5, 8, 9]. LSTM and hybrid recurrent networks are well established for sequential inertial biometrics [10, 11, 12]. Recent work has also studied smartphone authentication from gyroscope and accelerometer data for mobile educational resources [13]. That study uses correlation-based decisions rather than recurrent classification or online threshold adaptation and therefore does not establish the performance of the present method. A recent systematic review synthesizes 80 behavioral-biometric CA studies and maps security, privacy, intelligent-agent, and deployment gaps [14]. Beyond CA, members of the author team have applied machine learning to financial-feature analysis [15, 16, 17], medical imaging [18], and UAV recognition [19]. These domain-specific studies provide broader feature-engineering and deployment context; they are not used as evidence for authentication performance. Exponentially weighted statistics are likewise standard tools for tracking non-stationary streams [20, 21].

Table 1. Positioning of the present evaluation. “Online update” means a label-blind state change during the chronological test stream; “drift test” means an explicit time-varying shift evaluated against a frozen operating point. A check mark indicates that the cited work meets the stated definition. Protocols and datasets differ, so error rates are not directly comparable.

Work	Inertial	Sequence model	Online update	Drift test
Sitová et al. [3]	✓	–	–	–
AUToSen [9]	✓	✓	–	–
Verma et al. [5]	✓	–	–	–
This study	✓	✓	✓	✓

The methodological contribution is a paired evaluation: frozen and EMA-updated LSTM thresholds start from the same validation profile and process identical ordered test streams within each drift condition. This isolates the final-stage update from classifier performance.

3. Dataset and Leakage-Resistant Protocol

3.1. HMOG data and parsing

HMOG nominally contains 24 smartphone-interaction sessions for each of 100 participants, including sitting and walking tasks [3]. We use the accelerometer and gyroscope. An HMOG inertial CSV row contains metadata before the

three physical axes; consequently, only columns 4–6 are read as x, y, z. This detail is leakage-critical because a session identifier must never enter the feature vector. Accelerometer and gyroscope samples are paired by acquisition order after truncation to their common length; no timestamp resampling is used. Six sessions containing an empty inertial file for one subject are skipped, and session 24 is absent for two other subjects; all retain enough valid sessions to appear in every partition.

At the nominal 100-Hz sampling rate, the first five seconds are discarded. To equalize session contributions and bound computation, the next 20 non-overlapping windows of 100 samples are retained. Each window is represented by mean, standard deviation, minimum, maximum, root mean square, and skewness for each of six channels, producing 36 features. Ten adjacent windows form a 10-second sequence. The sequence step is also ten windows, so evaluation endpoints do not share raw samples. Sequences never cross a session boundary. The procedure yields 4,784 sequences from 2,392 valid sessions.

3.2. Three-way session split

For session number r, define $g(r) = (r - 1) \bmod 5$. In outer fold f, sessions with $g(r) = f$ are test sessions, sessions with $g(r) = (f + 1) \bmod 5$ are validation/calibration sessions, and the remaining sessions are used to fit the model. Whole sessions, rather than sequences, are assigned to partitions. Every included subject is required to occur in all three partitions. This is a session-disjoint cross-validation protocol, not a claim of leave-one-subject-out or prospective unseen-subject recognition.

A standard scaler is fitted only to the model-fitting sequences and then applied unchanged to validation and test sequences. Early stopping and initial threshold estimation use validation data; test labels are not used for either. The same validation fold is used for checkpoint selection and claimant-specific threshold/profile initialization; it is test-disjoint but not an independent calibration set. This is closed-set verification over 100 enrolled identities. For claimant u, the class-u softmax probability is the score; sequences from u are genuine and those from the other 99 identities are zero-effort impostors. All identities occur during fitting; unseen impostors are not tested.

4. Method

4.1. System overview

The system has five stages:

sensor samples → 1-s windows → 36 features → LSTM score → frozen or EMA-updated threshold

The first four stages are identical in the ablation. Only the last-stage threshold update changes. Within each paired condition, FAR/FRR differences therefore reflect the update rule rather than a different classifier or window.

4.2. LSTM classifier and baselines

The input is $\mathbf{X} \in \mathbb{R}^{10\times36}$. A two-layer LSTM with 64 hidden units per layer returns its last hidden state. Layer normalization, a 32-unit ReLU layer, dropout of 0.3, and a 100-class softmax produce claimant scores $s_{u,n} \in [0, 1]$. Training uses cross-entropy and Adam with learning rate 10^{-3}, batch size 256, a maximum of 20 epochs, and early-stopping patience 4. Here "agent" denotes only the stateful EMA decision rule described below; it does not retrain the LSTM or perform planning.

Linear SVM and random forest baselines receive the same sequence flattened to 360 dimensions. Linear SVM uses $C = 0.1$; the forest contains 300 unrestricted-depth trees, uses square-root feature subsampling, and has minimum leaf size 2. These fixed reference baselines compare classifier families at the same temporal endpoint; they are neither tuned as state-of-the-art comparators nor presented as adaptive systems.

4.3. EMA adaptive agent

For claimant u, the validation genuine scores initialize the first and second moments $p_{u,0}$ and $q_{u,0}$. Before attempt n, the agent computes

$$\sigma_{u,n-1} = \sqrt{\max(q_{u,n-1} - p_{u,n-1}^2, \epsilon^2)}, \tag{1}$$

$$\tau_{u,n} = \max(0, p_{u,n-1} - k_u \sigma_{u,n-1}), \tag{2}$$

$$d_{u,n} = \mathbb{I}[s_{u,n} \geq \tau_{u,n}]. \tag{3}$$

When $d_{u,n} = 1$, the accepted score updates both moments:

$$p_{u,n} = (1 - \alpha) p_{u,n-1} + \alpha s_{u,n}, \tag{4}$$

$$q_{u,n} = (1 - \alpha) q_{u,n-1} + \alpha s_{u,n}^2. \tag{5}$$

Here $\epsilon = 10^{-4}$ supplies a strictly positive scale floor when validation-score variance is negligible. After a rejection the state is unchanged. Thus repeated rejected events cannot lower the threshold. An accepted impostor can still poison the profile; the evaluation deliberately preserves that risk instead of using ground-truth labels to gate updates.

We fix $\alpha = 0.05$ a priori, corresponding to a characteristic update horizon of approximately $1/\alpha = 20$ accepted scores. For each claimant, the validation genuine and impostor scores determine an EER threshold $\tau_{u,0}$; the corresponding $k_u = (p_{u,0} - \tau_{u,0})/\sigma_{u,0}$ makes the first adaptive threshold exactly equal to that test-independent, validation-calibrated operating point. Thus k_u is not selected on test scores. The frozen LSTM ablation uses the same $\tau_{u,0}$ but never updates it. Decisions use the current claimant score; no additional score buffer is used.

5. Evaluation

For each claimant, EER is approximated at the held-out ROC point minimizing |FAR − FRR|. Claimant values are macro-averaged within each fold, after which mean and standard deviation are computed over five folds. This test-derived point summarizes separability and is never used for decisions.

The frozen-threshold and EMA-agent decisions start from the same validation-calibrated operating point. Genuine test attempts follow session and within-session chronology. Each is alternated with a zero-effort impostor score sampled without replacement uniformly from all test sequences of the other 99 identities; sampling is repeated ten times, with the validation profile reinitialized before each repeat. Genuine chronology is unchanged while the sampled impostor stream varies. This balanced 50% impostor stream is an explicit stress test, not an estimate of real attack prevalence. Because the agent sees no labels, an accepted impostor updates its profile. We report macro false acceptance rate (FAR), false rejection rate (FRR), and HTER = (FAR + FRR)/2.

To study score drift, one unit direction sampled from $\mathcal{N}(0, I_{36})$ with seed $42 + f$ is ramped linearly over each subject's test chronology to final ℓ_2 magnitudes $\|\delta\|_2 \in \{0, 1, 2\}$. At each endpoint, the same 36-dimensional shift is added to all ten time steps of a sequence; the stated norm refers to the per-time-step standardized feature vector. The transformation is applied to all test samples, including samples subsequently drawn as impostors. The same trained model is used. This is a synthetic feature-shift sensitivity analysis, not a model of natural behavioral drift; only frozen-versus-EMA comparisons within each level are paired. Impostor sampling uses seed $42 + 100f + 10\|\delta\|_2$.

6. Results

Table 2. Macro EER on undrifted held-out sessions (mean ± standard deviation over five outer folds). All methods use the same 10-second sequences.

Model	EER (%)
Linear SVM	29.92 ± 0.80
Random forest	24.05 ± 0.86
LSTM (softmax score)	21.87 ± 1.01

Table 2 compares score separability. Table 3 isolates the agent at a threshold estimated without test data. The agent is not assigned an additional post-hoc EER; its chronological, label-blind decisions are reported.

Table 3. Frozen-threshold and EMA-agent operating points in the balanced alternating stream (mean ± standard deviation over five folds).

$\|\delta\|_2$	Decision rule	FAR (%)	FRR (%)	HTER (%)
0	Frozen	22.55 ± 1.13	23.82 ± 2.27	23.19 ± 0.95
0	EMA agent	24.46 ± 2.12	24.60 ± 2.04	24.53 ± 1.28
1	Frozen	22.43 ± 0.69	24.82 ± 1.97	23.63 ± 1.10
1	EMA agent	24.30 ± 2.00	25.73 ± 2.64	25.02 ± 1.66
2	Frozen	22.40 ± 0.85	27.37 ± 2.91	24.89 ± 1.37
2	EMA agent	24.53 ± 2.11	28.25 ± 4.25	26.39 ± 2.12

On undrifted data, the LSTM lowers macro EER by 2.18 percentage points relative to the random forest and by 8.05 points relative to the linear SVM.

The agent ablation is negative. Adaptation increases HTER by 1.34, 1.39, and 1.51 points at drift magnitudes 0, 1, and 2, respectively, using unrounded fold means. The frozen threshold has lower HTER in all five folds at every level. Both FAR and FRR rise on average, so this is not a deliberate security–usability trade-off. Observed HTER is higher at larger nominal drift, although independent impostor resampling also contributes to cross-level variation. These fold summaries are descriptive: training and validation partitions overlap across outer folds, so the reported standard deviations are not confidence intervals and no statistical-significance claim is made.

7. Discussion

Table 2 measures discrimination, whereas Table 3 evaluates the test-independent, validation-calibrated operating point. Low EER alone does not establish an adaptation benefit.

The label-blind accepted-only update cannot lower the threshold after a rejection, cannot adapt to sub-threshold genuine scores, and remains vulnerable to poisoning after a false acceptance. Future deployment studies should therefore evaluate high-confidence update gates, step-up authentication, and poisoning defenses.

7.1. Limitations

Evaluation is limited to HMOG; six empty-file sessions and two absent directories are excluded. Row-order sensor pairing may leave temporal misalignment, and the 20-second excerpt may emphasize early-session behavior. The modulo split is not a rolling-origin forecast, produces unequal fold sizes, and is not explicitly stratified by activity or context. The alternating 50% impostor stream fixes genuine before impostor and does not model attack prevalence or bursts. Independent impostor draws at each drift level confound cross-level changes with Monte Carlo variation; only frozen-versus-EMA comparisons within a level are paired. Drift uses one direction per fold. Only one prespecified EMA rate, $\alpha = 0.05$, is evaluated; the negative result does not generalize to other update rates or adaptation rules. Future work requires independent datasets, natural sensor shifts, confidence intervals accounting for subject/session clustering, and on-device latency and energy tests.

Data and code availability. Code, protocol metadata, and fold results are archived in the public GitHub release v1.0.0. HMOG data are not redistributed.

8. Conclusion

Under this session-disjoint HMOG protocol, the LSTM has the lowest undrifted macro EER (21.87% ± 1.01%), but EMA adaptation raises HTER from 23.19% to 24.53% without drift and from 24.89% to 26.39% at drift magnitude 2. The tested rule does not improve the validation-calibrated frozen threshold; independent datasets, natural drift, poisoning, and on-device tests remain necessary.

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